Palaeomagnetism of the Younger Gabbros of Aberdeenshire and its Bearing on their Deformation

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ABSTRACT: The directions of magnetisation of gabbros from all the large Younger Basic masses in Aberdeenshire are found to be essentially the same. Any post-consolidation folding of these masses, as proposed by some workers, is therefore unlikely to have occurred. Precautions considered to be necessary in the collection of samples of basic igneous rocks for palaeomagnetic investigations are discussed.

1. INTRODUCTION

AT THE BEGINNING of this century it was discovered that igneous rocks possessed a remanent magnetisation of great stability. This phenomenon was used by a number of workers in an attempt to retrace the magnetic field of the earth in geological time, chief amongst whom was Konigsberger. In 1938 he was the first to point out that the remanent magnetisation of an igneous rock was probably acquired when the rock cooled and that it was directed along the earth's field acting at that time. Roche (1950), working on successions of recent lavas in the Plateau Centrale, demonstrated that the flows are magnetised, on the average, not along the earth's present field but in the direction of a field due to a supposed magnetic dipole located at the centre of the earth and directed along its rotational axis. Hospers (1954) considerably strengthened this interpretation in his systematic research on recent Icelandic plateau lavas, when he found that the rocks were magnetised on the average in the direction of the geographic

North Pole with an inclination downwards appropriate to the axial dipole field.

As a result of a systematic investigation of the magnetic properties of the New Red Sandstones in England, Clegg, Almond & Stubbs (1954) were able to show that the rocks were magnetised along a NE.–SW. axis with a shallower inclination than the earth's present field. Creer, Irving & Runcorn (1954, 1957) have measured the directions of magnetisation of many British rocks of various ages. They have presented the palaeomagnetic data indicating the direction of magnetisation for rocks from Miocene to Pre-Cambrian and the inferred pole positions at a number of times since the deposition of the Lower Torridonian Sandstones of Scotland. It is clear that these results have an important bearing not only on the theory of continental drift but also on the possibility of more humble movements, one of which is the concern of this paper.

The usefulness of rock magnetism in the solution of any geological problem is governed by the need to demonstrate that the magnetisation measured in the laboratory is a true record of that acquired by the rock when it was formed. In igneous rocks, the magnetisation can in appropriate cases be shown to be of thermo-remanent origin, acquired when the rocks cooled from above the Curie point of the ferromagnetic minerals concerned. Any subsequent metamorphism would naturally modify this magnetisation. A number of stability criteria, by which the constancy of direction of magnetisation of a rock can be demonstrated, are fortunately available.

Blundell (1957b) has utilised the magnetic properties of rocks to investigate a number of geological problems in order to illustrate the use of palaeomagnetism in geology. Among these problems was that concerning the possible deformation of the large gabbro bodies in north-east Scotland, investigated in the main by Read. This account is put together not only for its relevance to this particular question but also to direct attention to a variety of precautions that seem to be necessary in collecting material for palaeomagnetic study.

2. EXPERIMENTAL PROCEDURE

Orientated samples were collected large enough to provide at least two cube specimens of two centimetre side; the distinction between sample and specimen is the general practice and is followed in this paper. Before removal from the exposure, the sample was marked on its upper surface with an arrow directed towards the magnetic north as observed by using a prismatic compass. Horizontal lines were marked on two adjacent vertical faces of the sample, preferably at right angles to one another, by use of a spirit-level. With this technique samples were correctly orientated to within three degrees.

From the samples, cubes of two centimetre side were prepared by first cutting a two centimetre thick slice parallel to the horizontal markings, transferring the arrow markings to this slice and then cutting the final four faces. By this procedure, cubes accurate to within half a millimetre could be quickly obtained. Errors in orientation of the cubes with reference to the *in situ* orientation in no case exceeded 5°. Since there was statistical variation in directions of magnetisation in the rocks investigated greater than this, there was no need for a more accurate technique. Both Robertson (1947) and Graham (1949) have shown that rocks possessing stable magnetisation are unaffected in the process of removal of the samples and the preparation of the cubes therefrom.

Blundell (1957a) has described the experimental procedure: 'Before measurement of the magnetisation of the specimens was made, the samples were "stored for a month in the present earth's field orientated as they occurred *in situ*. After measurement, each one was turned through 180° about an E.–W. axis, to lie for another month, and was then re-measured. This process served two purposes: first, the mean of both measurements removed isothermal and viscous magnetisation, leaving only the natural remanent magnetisation which was derived from the original thermo-remanent magnetisation acquired during the formation of the (rock); second, it is a standard method of testing the stability of the remanent magnetisation, since those specimens whose direction of magnetisation altered greatly between the two observations are probably unstable. A further test of stability is afforded by the calculation of the value Q_N, the ratio of natural remanent magnetisation to induced magnetisation." Dutt (1955) showed that for the Mull lava flows, the greater the value of Q_N the greater is the stability of magnetisation' (Blundell, 1957a, 188–9). If the value of Q_N is greater than unity, the magnetism of an igneous rock is clearly of thermo-remanent origin, since it is unlikely that the earth's field was ever much greater than it is today.

The magnetisation was measured on a spinning magnetometer designed by Bruckshaw & Robertson (1948) and modified by Vincenz (1952). The cubes were rotated about all three axes, thus allowing a check on the accuracy of the observations. The initial susceptibility, employed in the determination of the quantity Q_N already mentioned, was measured on an instrument built by Bruckshaw & Robertson on the same principle as their spinning magnetometer.

3. THE SUPPOSED DEFORMATION OF THE YOUNGER GABBROS OF NORTH-EAST SCOTLAND

The Younger Basic intrusions of Aberdeenshire form one of the most distinctive igneous provinces of Scotland. In date, they are later than the

regional metamorphism of the Dalradian country-rocks but earlier than the deposition of the local Middle Old Red Sandstone. They make half a dozen large and many small bodies of igneous rock that are united by such provincial characters as their dominant gabbroic composition, the occurrence of peridotitic and troctolitic derivatives often banded, the abundance of noritic types and the common development of contaminated varieties. The chief intrusions of the province are shown in Fig. 1. Since 1914, many of them have been studied in whole or in part and a considerable body of knowledge is accumulating about them (Watt, 1914; Read, 1921, 1923a, 1923b, 1924, 1931, 1935, 1951, 1952, 1956; Read & Farquhar, 1952; Stewart, 1946; Whittle, 1936; Sadashivaiah, 1950, 1954a, 1954b; Shackleton, 1948). There can be little doubt that all these basic bodies belong to one igneous event and possibly form together a great sill-like body miles thick and hundreds of square miles in lateral extent.

The Dalradian country-rocks of these intrusions are metamorphic rocks, originally sediments, varying in grade from slate to regional gneisses of different types and origins. In appropriate situations, the country-rocks are thermally contact-metamorphosed by the gabbros with the production of normal hornfelses of static types. According to Read (1955) the metamorphosed Dalradian rocks are arranged in a gigantic recumbent anticline closing south-eastwards and itself affected by subsidiary broad structures, one of which is the Buchan Anticline (Read & Farquhar, 1956). Resting unconformably on the Dalradian are outliers of Old Red Sandstone sediments which contain material derived from the Younger Basic suite (Read, 1923b; Hinxman & Wilson, 1890). The western margins of some of these outliers are formed by faults and their rocks are broadly folded (Read, 1923b).

In three of the basic masses, Belhelvie, Huntly and Insch, banded and other structures occur, some of which have been used to propose an overturning of the gabbro masses since their consolidation. In the Belhelvie mass, Stewart (1946) observed rhythmic banding along the western part of the mass in troctolite, hypersthene-gabbro and western olivine-gabbro. The banding is considered to be due to the accumulation under the influence of gravity of the early formed crystals during the consolidation of the magma and to be originally more or less horizontal in attitude. Stewart favours the opinion that the banded rocks have undergone a considerable post-banding tilting, even though the floor of the chamber may have had an initial dip. Since the sedimentation characters of the Belhelvie bands indicate younging to the east, Stewart's opinion requires a 50° to 90° tilt from west to east about a north–south axis. In the peridotite and troctolite of the western part of the Huntly mass, Shackleton (1948) has shown that the rhythmic banding strikes north and south and dips westwards at 70°. On sedimentation characters acquired during the settling of the heavy

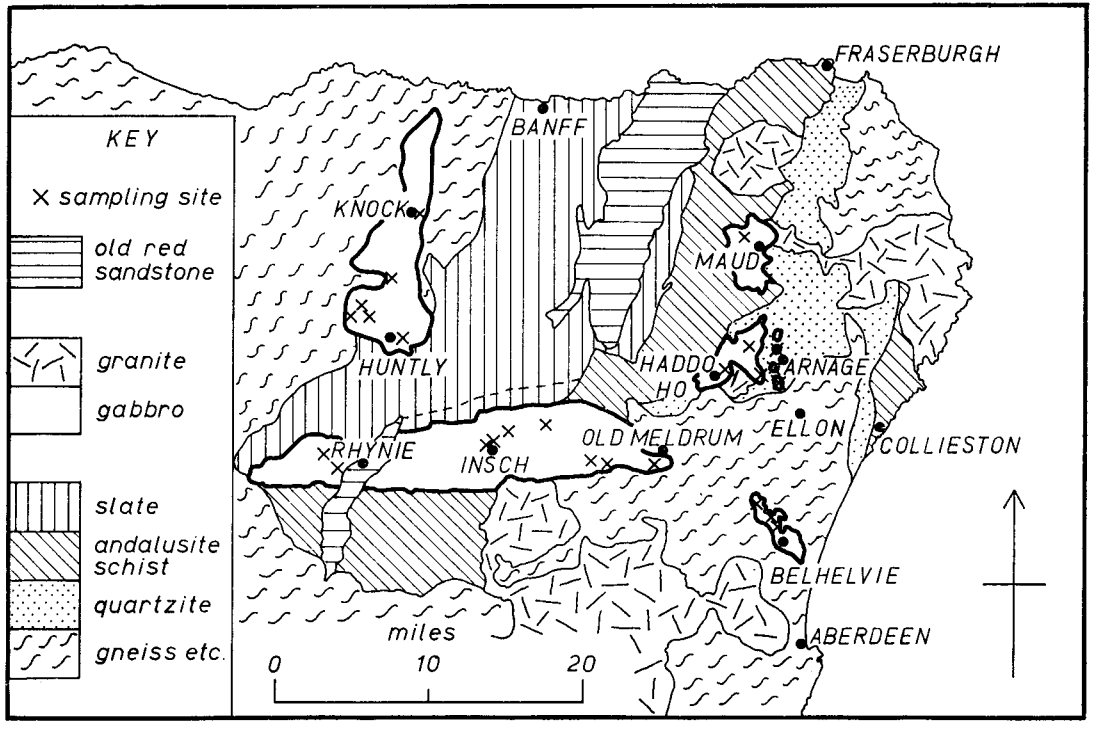


Fig. 1. The Basic Masses of North-East Scotland, with sampling sites of investigated specimens

crystals, Shackleton considers that the banded rocks of the Huntly mass young or face eastwards and are thus overturned; he concludes that this overturning took place when the gabbros were rigid and is thus a tectonic phenomenon. For the Belhelvie and Huntly masses, therefore, tilts of 90° or more from west to east about a north and south axis are proposed after the consolidation of the gabbro magma.

The Insch banded gabbros do not supply evidence of the quality of that from Belhelvie and Huntly. In the eastern end of the Insch Mass, Sadashivaiah (1954a) described peridotites, troctolites and olivine-gabbros, that from their limited outcrops available seemed to run in north–south bands and, in some exposures, small-scale banding that appears to be steep can be seen. Other evidence, not yet published, derived from the attitude of the large-scale differentiates of the olivine-gabbro of the Insch mass, shows that the original uppermost part of the Insch differentiated mass is still uppermost—no tilting in the Insch mass as a whole can have taken place.

These data have been assembled by Shackleton (discussion to Read & Farquhar, 1956) to form Fig. 2. From them, he deduces a large-scale post-gabbro folding responsible for the overturning of the originally flat banding of the Huntly and Belhelvie masses with, of necessity, the formation of the Buchan Anticline in the Dalradian rocks and the upending of the pile of

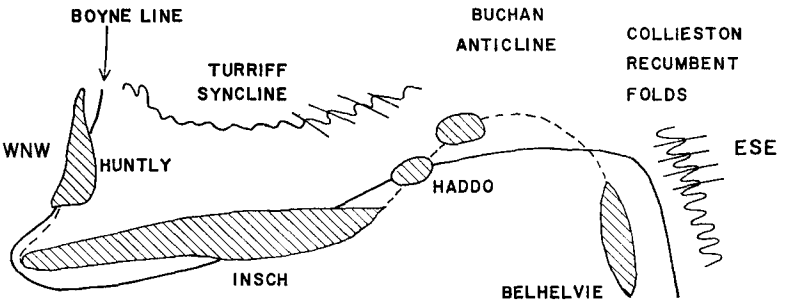


Fig. 2. Professor Shackleton's interpretation of the attitude of the Basic Masses of North-East Scotland. (Reproduced by permission of the Council of the Geological Society from *Quart. J. geol. Soc. Lond.*, **112**, 115)

small-scale recumbent folds of Collieston (see Fig. 2). Read (discussion to Read & Farquhar, 1956) was unable to accept this post-consolidation folding of the basic masses. There was no doubt that the Dalradian country-rocks had undergone as part of their major structure a small-scale tight folding that was earlier than the regional metamorphism, this being of a static post-kinematic type characterised by andalusite and cordierite. Shackleton (discussion to Stewart, 1946, p. 498, and 1948) had suggested that the change of attitude was largely achieved by distortion. The gabbros

of this province abound in delicate textures such as reaction rims, dactylitic intergrowths and the like, and intricate textures are present in the abundant contact-hornfelses and the country-rocks generally. The perfect preservation of such textures indicated to Read (Read & Farquhar, 1956) that no post-consolidation overturning, achieved by distortion, had occurred. Further, he found it difficult to accept the separation of the broad 'Buchan Anticline' folding from the main recumbent folding of the Banff Nappe by a considerable interval during which the Younger Basic gabbros were intruded. It may be recalled that the Middle Old Red Sandstone of north-east Scotland shows in places a degree of folding. This post-Old Red Sandstone folding appears to Read to be of a shallow type unlikely to have affected the rigid gabbros and basement on which the outliers lie. In 1958, Shackleton reaffirmed his opinion that the Banff Nappe and the basic sheet intruded in it had been deformed by movements which produced steeply monoclinal zones (1958, 392).

The controversy outlined above is clearly one that might be settled by the application of rock magnetism. The Younger Basic rocks may be considered to be of Caledonian origin and probably of Lower Palaeozoic age. Creer *et al.* (1954) had measured the direction of magnetisation of the Cambrian Caerbwdy Sandstone of South Wales and found it to be directed southwards with 40° downward inclination. If the Aberdeenshire gabbro masses were in their original positions relative to one another, they should all have this direction of magnetisation, but if any masses had been turned about a NNE.–SSW. axis through about 100° as indicated in Shackleton's diagram, such masses would have a direction 60° W. of S. and downward inclination of 10°. These directions are 65° apart. Thus it was considered that the two possibilities might be resolved. If Shackleton's hypothesis was correct, then samples from Huntly and Belhelvie would be magnetised in a direction 60° from that of samples from the Insch, Haddo House and Arnage masses; if Read was correct all samples should be magnetised southerly with 40° downward inclination. It was of course realised that if, at the time the gabbro magma consolidated, the earth's field was directed along the axis of the postulated folding, the method of testing would be invalid, since no change of direction of magnetisation would have occurred. A further requirement was the demonstration that the magnetisation was acquired by the rocks during consolidation and had remained unaltered in direction up to the present time.

4. THE SAMPLING AND THE GEOPHYSICAL AND GEOLOGICAL CONTROLS OF THE VALID SPECIMENS

It will be recalled that samples are the orientated pieces of rock collected at a site and specimens are the individual cubes or cylinders cut from the

samples. The sampling sites in the different masses are indicated in Fig. 1. Twenty-six sites yielded sixty samples which were examined petrographically and geophysically to determine their suitability for palaeomagnetic purposes. The sites and the controls governing the selection of specimens from them are now detailed.

(a) The Huntly Mass

In this mass, samples, totalling fourteen in number, were collected from Bin Hill Quarry, three miles NW. of Huntly, in banded olivine-gabbro; from Sinsharnie Quarry, one half-mile N. of Bin Hill, in troctolite and peridotite, the latter mostly serpentinised; from Craighead Quarry, one mile W. of Huntly, in partly serpentinised peridotite; from Battlehill Quarry, Huntly, in gabbro; from Rothiemay Railway Cutting in heavily serpentinised peridotite, and from the great Ternemny Quarry near Knock, eight miles N. of Huntly, in fresh biotite-norite. The rock in Battlehill Quarry is clearly greatly deformed and fractured and in slice proved to be partly of contaminated origin (cf. Read, 1923b)—the samples were rejected for palaeomagnetic purposes.

(b) The Insch and Cabrach Masses

These masses can be considered as originally one, now separated by the post-Middle Old Red Sandstone faulting in the Bogie Valley. Eighteen samples were collected from eight sites in these masses. The largest collection came from Pitscurry Quarry near Pitcaple and from Cuttlecraigs Quarry, a mile E. of Pitscurry. These rocks are gabbros, norites and granular gabbros, often greatly deformed and in places essentially mylonitic. Read (1956) has shown that the south-western margin of the Insch Mass is formed by a shear zone along which the country-rocks in the south have moved upwards relative to the igneous body and that similar shear-belts traverse the gabbros in the Insch and Ledikins areas (Read, 1956). Further, nearer Pitscurry and Cuttlecraigs, the same author (Read, 1951) has described the production of cataclastic and mylonitic rocks in weak elements in the gabbros and country-rocks of the Pitcaple area. When examined for their magnetisation, the samples from Pitscurry and Cuttlecraigs showed a wide variation in their inclination, possibly not resulting entirely from instability but from disorientation brought about by the shear movements. The upwarping of the southern boundary of the Insch Mass could account for the low inclination values observed. All the samples from the two quarries were finally rejected.

A sample from Fallow Hill, though having a magnetisation apparently stable and in agreement with other stably magnetised samples, was rejected on account of decomposition due to weathering.

(c) The Belhelvie Mass

Three sites were sampled in this mass (Stewart, 1946), namely, Craigie Quarry, Beauty Hill Quarry and large solid outcrops on Harestone Moss. These provided olivine-gabbro, troctolite and peridotite, the more ultrabasic types being partly serpentinised but showing no signs of mechanical deformation.

(d) The Haddo House Mass

Collections from the Haddo House mass illustrated the hazards of sampling gabbros for palaeomagnetic purposes. Although ten samples were collected, only one was finally considered to be truly *in situ*. This was the biotite-norite in contact with calcareous country-rock exposed in the quarry at Michael Muir on the east edge of the Haddo House body (Read, 1935). Much of the Haddo House mass is exposed as residual boulders, satisfactory it is believed for mapping purposes but useless for palaeomagnetic studies. These boulders vary in size up to forty feet in length but are generally only about ten feet long. It can be demonstrated in some places that what looked like a good *in situ* outcrop is really, when uncovered, a group of large boulders only roughly in place. Such boulders are in place to the extent that they have probably not moved more than a few inches or feet from their original positions, but in doing so they have of course become disorientated. The evolution of the residual boulders can be seen to be controlled by the characteristic exfoliation-weathering of the gabbros. As the various layers peel off, a residual core of fresh massive gabbro is left sitting in the soft, fragmented, weathered material. Settling and rotation could then take place relatively easily and in this manner very large boulders become disorientated. Even in seemingly solid unmoved crags, movement on widened joint surfaces is possible. The magnetic measurements of the Haddo House boulders appear to indicate a rotation about a vertical rather than a horizontal axis, possibly due to the prolate shape of the residuals.

(e) The Arnage Masses

In the small Arnage masses, nine samples were collected from exposures in cuttings of the Ellon to Fraserburgh railway line. The rocks from three sites were not truly magmatic material and from two others were highly contaminated (Read, 1923a; Read & Farquhar, 1952). The four pure magmatic samples came from Little Arnage, Arnage Station and Towie Wood and were norites as described by Read (1923a). Although not specified in Shackleton's original hypothesis, the Maud mass would presumably be placed in much the same position as the Haddo House and Arnage masses. Two samples, a quartz-gabbro and a norite, were obtained from a quarry and a crag near Mill of Fedderate, a mile north-west of Maud; there was little doubt that the crag was solidly in place.

The intensity and direction of magnetisation and the susceptibility of the specimens were measured in the usual way and stability tests were carried out. Most samples were magnetised at right angles to the direction of the present earth's field, which gives evidence for some stability. As has been mentioned, there was a general tendency for rocks in which deformation had occurred to be unstable. The same appeared to be true for samples affected by end-stage processes, the low-temperature operation in which pyroxenes have been altered to colourless hornblendes arranged in clusters of small blades (Read, 1935). High-temperature reactions had no obvious effect on the magnetic stability. No correlation could be found between degree of serpentinisation of the peridotite samples and their stability of magnetisation—of the ten samples collected four were unstable. These observations on the relationship of stability to petrogenetic history are of course tentative, but they might be borne in mind in future work on palaeomagnetism in basic igneous rocks. Nine samples, all of them unaltered and undeformed gabbros, norites or peridotites, were found to be unstable to varying extents. Finally, twenty-one samples of fresh magmatic rock, apart from serpentinisation in peridotite, satisfied the stability criteria and were used in the palaeomagnetic tests.

5. THE MAGNETIC RESULTS

The general magnetic results for the stable rocks are summarised in the Table below. In Fig. 3 the directions of magnetisation of the stable specimens and the mean directions for each mass are shown. The single useful sample from Haddo House has been grouped with those from Arnage. Since the two masses are close together, with no suggestion of relative movement, this is considered to be legitimate.

Mass	No. of samples	Azimuth ø mean	Inclination I mean	Konigsberger Ratio Q_N mean	Circle of confidence, radius	Distribution Factor K
Huntly	6	192° E. of N.	+60° down	2.1	8°	20
Insch	5	176	+48	0.7	14°	10
Arnage	4	186	+39	18.4	12°	13
Haddo House	1					
Belhelvie	3	174	+56	2.1	8°	24
Maud	2	177	+50	2.1		
All samples	21	182	+51	5.3	5°	13

It will be noted that values of Q_N for the norite samples from Arnage were considerably higher than those for any of the others. Nagata (1933, 125) has reported finding two rock samples from the same gabbro having the same susceptibility but differing by a factor of twenty in intensity of

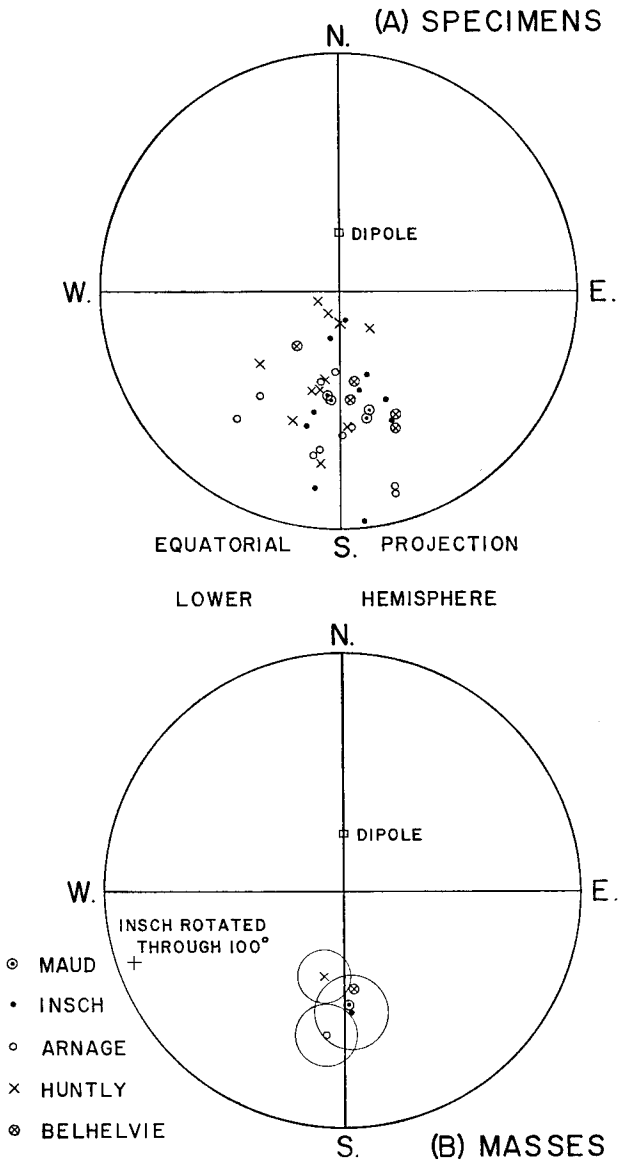


Fig. 3. Schmidt Equal Area Projections showing the Directions of Remanent Magnetism, (a) stable specimens, (b) mean directions for the Masses with circles of confidence

magnetisation. He found that the strongly magnetised rock contained both maghemite and magnetite, the weakly magnetised one only magnetite. Whether this holds for the Arnage rocks remains to be investigated.

The statistical analysis proposed by Fisher (1953) has been used to obtain the circles of confidence, radius α, calculated at the 33% probability level using samples as statistical units, for the Huntly, Insch and Arnage (with Haddo House) masses. The results are plotted in Fig. 3. The distribution factor K has also been calculated for each of the masses.

With the use of Watson's (1956) extension of Fisher's statistical analysis, a test was made of the hypothesis that the mean directions of magnetisation of the three sample populations from Huntly, Insch and Arnage–Haddo House were identical. Watson has shown that for three populations

$$\{\Sigma(R_i - R)/(N - \Sigma R_i)\} \cdot \{2(N-3)/4\} = F_{4,z(N-3)}$$

where N denotes the total number of samples, R the vectorial sum of all the magnetisation directions and R_i the vectorial sum of the directions in the ith population. $F_{4,z(N-3)}$ is the statistic of the F test for equality of variances with minimum and maximum degrees of freedom of 4 and z(N–3).

The calculated value of $F_{4,26}$ was 1.50, significantly less than the tabulated value at the 5% probability level of 5.76, thus indicating that there is no significant difference between the mean directions of the remanent magnetisation of the three sets of samples.

6. CONCLUSIONS

On the assumption that the Insch mass remains horizontal, then the Huntly mass, if overturned according to Shackleton's hypothesis, should have a magnetisation 70° from the direction actually observed in samples from Huntly. The possibility that such a difference is fortuitous is very small, being less than one per cent. The magnetisation of samples from Belhelvie is in full agreement with that from the other masses. Since the samples used in the statistical analyses were all of unaltered undeformed magmatic material, there is little likelihood that the rocks acquired their magnetisation after any supposed folding. The Q_N values have indicated that the magnetisation was thermo-remanent in origin and there was no sign of the samples having suffered any thermal metamorphism subsequent to their original intrusion and consolidation.

It must be concluded from this evidence that the folding suggested by Stewart and Shackleton is most unlikely to have occurred. There is no significant difference between the magnetisation of any of the basic masses, and it is therefore suggested that they have remained relatively undisturbed since their formation. The mean distribution of magnetisation of all the

stably magnetised samples is in general agreement with Creer's value for the Caerbwdy Beds and congruent with the intrusion of the Aberdeenshire gabbros as part of the Caledonian drama.

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